

Viva Questions for All Units (Simple)

UNIT I – Introduction to Spark

1. What are the limitations of MapReduce?
2. What is the difference between batch and real-time analytics?
3. What is stream processing?
4. What is in-memory processing in Spark?
5. What are the main features of Apache Spark?
6. Why is Spark faster than Hadoop?
7. How do you install Spark as a standalone user?
8. Compare Spark vs Hadoop ecosystem.

UNIT II – Introduction to Scala

1. What are the features of Scala?
2. What are Scala basic data types?
3. What are literals in Scala?
4. What is type inference?
5. Difference between mutable vs immutable collections.
6. What is a function in Scala?
7. What are lists and maps in Scala?
8. What are streams in Scala?
9. What are operators in Scala?

UNIT III – Using RDDs in Spark

1. What are the features of RDD?
2. How do you create RDDs?
3. What are RDD transformations?
4. What are RDD actions?

5. What is lazy evaluation?
6. What is the Spark shell?
7. What are shared variables?
8. Difference between broadcast and accumulator.

UNIT IV – Spark SQL

1. What is Spark SQL?
2. What are the features of Spark SQL?
3. How to convert RDD to DataFrame?
4. What is a DataFrame?
5. What is Hive integration in Spark?
6. What is a temporary view?
7. What is Catalyst Optimizer?

UNIT V – Kafka + Spark Streaming

1. What is Apache Kafka?
2. What is a Kafka topic?
3. What are Kafka partitions?
4. What is Kafka cluster architecture?
5. What is a Kafka broker?
6. What is a consumer group?
7. What is Spark Streaming?
8. What is real-time streaming?
9. How Spark connects with Kafka?
10. What is a DStream?
11. What is Kafka Java programming?

UNIT VI – Spark ML + GraphX

1. What is Machine Learning?
2. Difference between supervised and unsupervised ML.
3. What is Spark MLlib?
4. What are transformers?
5. What are estimators?
6. What is an ML pipeline?
7. Name some Spark ML algorithms.
8. What is GraphX?
9. What are vertices and edges?
10. What is feature extraction?
11. What is feature transformation?
12. What is frequent pattern mining?